Grade 11 – Term 1
C5, C6 and C7 Exam Solutions

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Learning objective

Use continuous probability models and sample data to describe typical values, variability, and interval probabilities, and interpret differences between sample estimates and theoretical values.

Criteria assessment

This exam assesses C5, C6 and C7 separately, with 24 points per criterion (72 points in total). Each task shows its point allocation. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. Answers without the required procedure will not receive credit. Failure to write your name on the task sheet or submitting it after the deadline will result in a one-point deduction from each criterion. For this exam, a minimum of 15 out of 24 points is required to pass each criterion. Use unrounded intermediate values; round final statistics to three decimal places and percentage errors to two decimal places where necessary.

Evaluated criteria

C5 Calculates and interprets the theoretical mean, variance, and standard deviation of uniform, triangular, linear, exponential, and normal distributions.

C6 Estimates interval probabilities using relative frequencies from samples and compares them with theoretical probabilities.

C7 Calculates sample means, variances, and standard deviations for different sample sizes and data representations and compares them with theoretical values.

1 C5: Theoretical Mean and Variance [C5-24]

1.1 Problem C5.1: Two packing stations [8]

Station A has a packing time A uniformly distributed between 4 and 16 minutes. Station B has a triangular packing time B with minimum 4 minutes, maximum 16 minutes, and mode 7 minutes. Use $\text{Tri}(a, b, c)$ with a the minimum, b the maximum, and c the mode.

Questions/Tasks

C5 Identify the model and calculate the mean, variance, and standard deviation for A. **[3]**

C5 Calculate the same three quantities for B. **[3]**

C5 Compare typical packing time and variability, including units. **[2]**

C5.1(a) – Uniform model [3]

The constant density on $[4, 16]$ gives $A \sim U(4, 16)$, so

$$E(A) = \frac{4+16}{2} = 10 \text{ min}, \qquad Var(A) = \frac{(16-4)^2}{12} = 12 \text{ min}^2, \qquad \sigma_A = \sqrt{12} \approx 3.464 \text{ min}.$$

C5.1(b) – Triangular model [3]

Here $B \sim \text{Tri}(4, 16, 7)$. Substitution gives

$$E(B) = \frac{4+16+7}{3} = 9 \text{ min},$$

$$Var(B) = \frac{4^2 + 16^2 + 7^2 - 4(16) - 4(7) - 16(7)}{18} = \frac{321 - 204}{18} = \frac{13}{2} = 6.5 \text{ min}^2.$$

$$\sigma_B = \sqrt{6.5} \approx 2.550 \text{ min}.$$

C5.1(c) – Interpretation [2]

B has a mean packing time one minute shorter than A and a smaller standard deviation ($2.550 < 3.464$ minutes), so it has less variable packing times. Variance is measured in square minutes; mean and standard deviation are in minutes.

1.2 Problem C5.2: Opposite linear models [8]

Two workshops have production times on $[2, 11]$ hours. For workshop I the density increases linearly from zero at 2 hours. For workshop D it decreases linearly to zero at 11 hours.

Questions/Tasks

C5 Identify each model and calculate its mean, variance, and standard deviation. **[6]**

C5 Explain why their means differ although their variances agree. **[2]**

C5.2(a) – Linear models [6]

The models are $I \sim \text{Lin}_{\text{inc}}(2, 11)$ and $D \sim \text{Lin}_{\text{dec}}(2, 11)$. Thus

$$E(I) = \frac{a+2b}{3} = \frac{2+2(11)}{3} = 8 \text{ h}, \qquad E(D) = \frac{2a+b}{3} = \frac{2(2)+11}{3} = 5 \text{ h}.$$

For either linear model,

$$Var(I) = Var(D) = \frac{(b-a)^2}{18} = \frac{9^2}{18} = 4.5 \text{ h}^2, \qquad \sigma_I = \sigma_D = \sqrt{4.5} \approx 2.121 \text{ h}.$$

C5.2(b) – Reasoning [2]

I places more probability near the upper endpoint, while D places more near the lower endpoint, shifting their means in opposite directions. The densities are reflections about 6.5 hours; reflection preserves squared distances from the respective means. Equivalently, both variance formulas depend only on the common interval width, 9 hours.

1.3 Problem C5.3: Arrivals and daily revenue [8]

Requests arrive according to a constant-rate Poisson process at 8 requests per hour. Let T be the time between consecutive requests in minutes. Daily revenue R, measured in hundreds of dollars, follows $R \sim N(24, 2.25)$, where the second parameter is the variance.

Questions/Tasks

C5 State the model for T in minutes and find its mean, variance, and standard deviation. **[4]**

C5 Find the mean, variance, and standard deviation of revenue in dollars. Interpret the mean. **[4]**

C5.3(a) – Exponential waiting time [4]

The rate in minutes is $\lambda = 8/60 = 2/15$ requests per minute. Consequently $T \sim \text{Exp}(2/15)$, and

$$E(T) = \frac{1}{\lambda} = \frac{15}{2} = 7.5 \text{ min}, \qquad Var(T) = \frac{1}{\lambda^2} = \frac{225}{4} = 56.25 \text{ min}^2, \qquad \sigma_T = \frac{1}{\lambda} = 7.5 \text{ min}.$$

C5.3(b) – Normal revenue and units [4]

For R, $\mu = 24$, $\sigma^2 = 2.25$, and $\sigma = \sqrt{2.25} = 1.5$ in hundreds of dollars. Let $Y = 100R$ be revenue in dollars. Then

$$E(Y) = 100(24) = 2400 \text{ dollars}, \qquad Var(Y) = 100^2(2.25) = 22500 \text{ dollars}^2,$$

$$\sigma_Y = \sqrt{22500} = 150 \text{ dollars.}$$

The model predicts a long-run average daily revenue of \$2400; individual days need not have that revenue.

2 C6: Empirical Probabilities [C6-24]

For all C6 problems use $\widehat{P} = k/n$, absolute error $E = |\widehat{P} - P|$, and percentage error $100E/P$ when requested.

2.1 Problem C6.1: Water consumption, raw data [6]

A simulated sample of daily household water consumption (hundreds of liters) is

2.1	2.5	2.9	3.0	3.2	3.6	3.9	4.0	4.3	4.7
2.4	2.8	3.1	3.4	3.7	4.0	4.1	4.5	4.8	5.2

The theoretical probability is $P(3 \le X \le 4) = 0.50$.

Questions/Tasks

C6 Find the empirical probability of the stated interval, showing the selected observations. **[3]**

C6 Calculate absolute and percentage errors and interpret the difference. **[3]**

C6.1(a) – Counting observations [3]

There are $n = 20$ observations. Those satisfying both inclusive endpoints are

$$3.0,\ 3.2,\ 3.6,\ 3.9,\ 4.0,\ 3.1,\ 3.4,\ 3.7,\ 4.0.$$

Thus $k = 9$ and $\widehat{P}(3 \le X \le 4) = 9/20 = 0.45$.

C6.1(b) – Errors [3]

$$E = |0.45 - 0.50| = 0.05, \qquad \text{Percentage error} = \frac{0.05}{0.50} \times 100\% = 10\%.$$

The empirical result is 0.05, or 5 percentage points, below the theoretical probability. The relative percentage error is 10%.

2.2 Problem C6.2: Service times, frequency data [6]

Recorded service times are summarized below.

x_i (min)	3	6	9	12	15	18
f_i	6	18	30	24	15	7

The theoretical probability is $P(6 \le X < 15) = 0.75$.

Questions/Tasks

C6 Find the sample size and empirical probability for this interval. **[3]**

C6 Find the absolute error and the empirical probability of the complementary event. Interpret the comparison. **[3]**

C6.2(a) – Frequency totals [3]

$$n = 6 + 18 + 30 + 24 + 15 + 7 = 100, \qquad k = 18 + 30 + 24 = 72.$$

The value 15 is excluded, so $\widehat{P}(6 \le X < 15) = 72/100 = 0.72$.

C6.2(b) – Complement and comparison [3]

$$E = |0.72 - 0.75| = 0.03, \qquad \widehat{P}(X < 6 \text{ or } X \ge 15) = 1 - 0.72 = 0.28.$$

The empirical interval probability is 3 percentage points below the model. The complementary counts $6 + 15 + 7 = 28$ confirm the complement result.

2.3 Problem C6.3: Download times, grouped data [6]

Two independent simulated samples from the same model give these frequencies. All times are in seconds; classes are left-inclusive and right-exclusive.

Class	$[0, 5)$	$[5, 10)$	$[10, 15)$	$[15, 20)$
Sample A	4	8	6	2
Sample B	18	78	76	28

The model gives $P(5 \le X < 15) = 0.80$.

Questions/Tasks

C6 Find each sample size, empirical interval probability, and absolute error. **[4]**

C6 Which estimate is closer? Does increasing sample size guarantee smaller error in every sample? **[2]**

C6.3(a) – Comparing samples [4]

The event contains the two middle classes, giving

$$n_A = 4 + 8 + 6 + 2 = 20, \quad k_A = 8 + 6 = 14, \quad \widehat{P}_A = 14/20 = 0.70, \quad E_A = |0.70 - 0.80| = 0.10.$$

$$n_B = 18 + 78 + 76 + 28 = 200, \quad k_B = 78 + 76 = 154,$$

$$\widehat{P}_B = 154/200 = 0.77, \qquad E_B = |0.77 - 0.80| = 0.03.$$

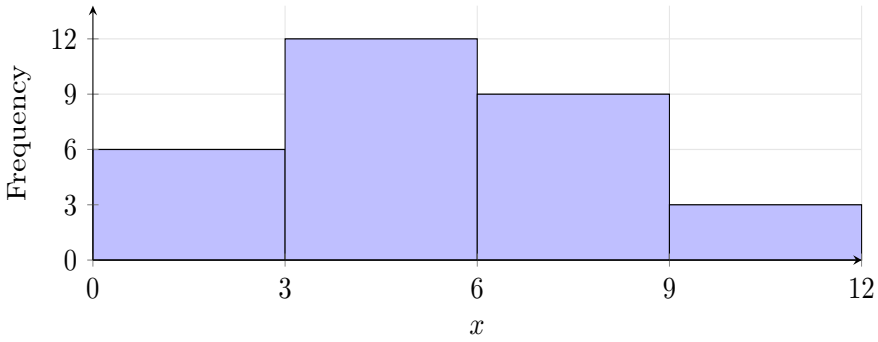
These relative frequencies are exact for the recorded samples because the event is a union of complete classes, although they estimate a model probability.

C6.3(b) – Sample size [2]

B is closer because $0.03 < 0.10$. Larger random samples generally produce more stable estimates, but random variation means that a larger sample need not have smaller error in every instance.

2.4 Problem C6.4: Ticket queues, histogram data [6]

The histogram is the only record of a sample of queue times. Bar heights are frequencies; times are in minutes, with left-inclusive, right-exclusive classes.



The theoretical probability is $P(3 \le X < 9) = 0.75$.

Questions/Tasks

C6 Read the frequencies and sample size. Find the interval probability and its absolute and percentage errors. **[3]**

C6 Give cumulative frequencies and empirical probabilities below each upper boundary, and find the probability of a queue time of at least 9 minutes. **[3]**

C6.4(a) – Reading the histogram [3]

The classes $[0,3), [3,6), [6,9), [9,12)$ have respective frequencies $6, 12, 9, 3$. Hence $n = 6 + 12 + 9 + 3 = 30$ and $k = 12 + 9 = 21$.

$$\widehat{P}(3 \le X < 9) = 21/30 = 0.70, \quad E = |0.70 - 0.75| = 0.05,$$

$$\text{Percentage error} = \frac{0.05}{0.75} \times 100\% = \frac{20}{3}\% \approx 6.67\%.$$

The estimate is 5 percentage points below the model probability.

C6.4(b) – Cumulative results [3]

Add frequencies successively and divide each running total by 30:

b (min)	3	6	9	12
Cumulative frequency	6	18	27	30
$\widehat{P}(X < b)$	0.20	0.60	0.90	1

Only the last class has times at least 9 minutes, so $\widehat{P}(X \ge 9) = 3/30 = 0.10 = 1 - 0.90$.

3 C7: Sample Mean and Variance [C7-24]

Use the sample-variance denominator $n-1$ throughout this section.

3.1 Problem C7.1: Sensor readings, raw data [6]

Five sensor readings in degrees Celsius are $9, 11, 12, 13, 15$. The theoretical model has $\mu = 12$ degrees Celsius and $\sigma^2 = 4$ square degrees Celsius.

Questions/Tasks

C7 Calculate the sample mean, sample variance, and sample standard deviation. **[4]**

C7 Compare the mean and variance with the model. Explain why dividing the squared-deviation sum by 5 does not give the requested sample variance. **[2]**

C7.1(a) – Raw-data statistics [4]

$$n = 5, \qquad \bar{x} = \frac{9+11+12+13+15}{5} = \frac{60}{5} = 12.$$

x_i	9	11	12	13	15
$x_i - \bar{x}$	-3	-1	0	1	3
$(x_i - \bar{x})^2$	9	1	0	1	9

The squared deviations total 20. Thus

$$s^2 = \frac{20}{5-1} = 5, \qquad s = \sqrt{5} \approx 2.236.$$

The mean and standard deviation are in degrees Celsius; the variance is in square degrees Celsius.

C7.1(b) – Model comparison [2]

The sample mean equals $\mu = 12$, but $s^2 = 5$ exceeds $\sigma^2 = 4$ by 1 square degree Celsius. Dividing by 5 gives the descriptive variance of these five values, $20/5 = 4$, rather than the requested sample estimator. Since the mean was estimated from the sample, this estimator uses $n - 1 = 4$.

3.2 Problem C7.2: Items sold, frequency data [6]

The numbers of items sold in six short sales periods are summarized below. The theoretical model has $\mu = 5$ items and $\sigma^2 = 5$ items^2.

x_i	2	5	8
f_i	1	3	2

Questions/Tasks

C7 Find the sample size, mean, variance, and standard deviation using frequencies. **[4]**

C7 Compare the sample mean and variance with the theoretical values. **[2]**

C7.2(a) – Weighted statistics [4]

$$n = 1 + 3 + 2 = 6, \qquad \bar{x} = \frac{1(2) + 3(5) + 2(8)}{6} = \frac{33}{6} = 5.5 \text{ items}.$$

The weighted squared deviations are

$$1(2 - 5.5)^2 = 12.25, \quad 3(5 - 5.5)^2 = 0.75, \quad 2(8 - 5.5)^2 = 12.5.$$

$$s^2 = \frac{12.25 + 0.75 + 12.5}{6 - 1} = \frac{25.5}{5} = 5.1 \text{ items}^2, \qquad s = \sqrt{5.1} \approx 2.258 \text{ items}.$$

C7.2(b) – Comparison [2]

The sample mean is $5.5 - 5 = 0.5$ items above the model mean; the sample variance is $5.1 - 5 = 0.1$ items^2 above the model variance. These small sample differences do not establish that the model is incorrect.

3.3 Problem C7.3: Cycling times, grouped data [6]

Cycling times in minutes are grouped as follows, using left-inclusive, right-exclusive classes. The theoretical mean is 18 minutes and variance is 20 minutes^2.

Time (min)	$[10, 14)$	$[14, 18)$	$[18, 22)$
f_i	1	2	3

Questions/Tasks

C7 Find midpoints and estimate the sample mean, variance, and standard deviation. **[4]**

C7 Compare with the model and explain why these statistics are estimates. **[2]**

C7.3(a) – Midpoint estimates [4]

The midpoints are $(10 + 14)/2 = 12$, $(14 + 18)/2 = 16$, and $(18 + 22)/2 = 20$. The sample size is $n = 1 + 2 + 3 = 6$.

$$\bar{x} \approx \frac{1(12) + 2(16) + 3(20)}{6} = \frac{104}{6} = \frac{52}{3} \approx 17.333 \text{ min}.$$

Using the unrounded midpoint mean, the squared-deviation sum is

$$1\left(12 - \frac{52}{3}\right)^2 + 2\left(16 - \frac{52}{3}\right)^2 + 3\left(20 - \frac{52}{3}\right)^2 = \frac{256 + 32 + 192}{9} = \frac{160}{3}.$$

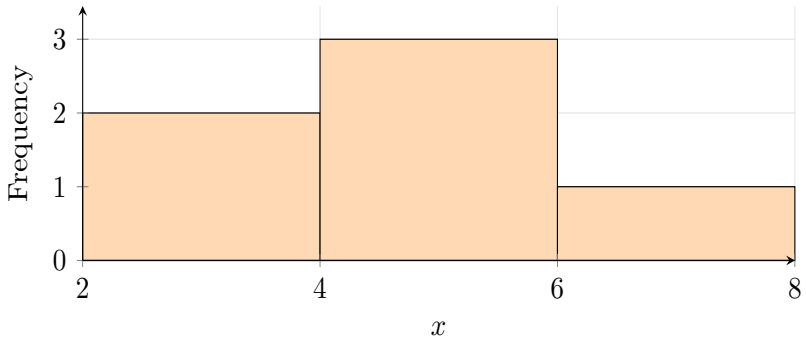
$$s^2 \approx \frac{160/3}{6-1} = \frac{32}{3} \approx 10.667 \text{ min}^2, \qquad s \approx \sqrt{\frac{32}{3}} \approx 3.266 \text{ min}.$$

C7.3(b) – Interpretation [2]

The estimated mean is $18 - 52/3 = 2/3 \approx 0.667$ minutes below the model mean. The estimated variance is $20 - 32/3 = 28/3 \approx 9.333$ minutes^2 below the model variance. Individual times are unknown and have been replaced by class midpoints, so these are estimates of the sample statistics.

3.4 Problem C7.4: Crate masses, histogram data [6]

A histogram gives a sample of crate masses in kilograms. Bar heights are frequencies; classes are left-inclusive and right-exclusive. The model has $\mu = 5$ kg and $\sigma^2 = 2$ kg^2.



Questions/Tasks

C7 Read the frequencies and sample size. Use midpoints to estimate the sample mean, variance, and standard deviation. **[4]**

C7 Compare with the model and state the limitation caused by grouping. **[2]**

C7.4(a) – Histogram statistics [4]

The classes $[2,4), [4,6), [6,8)$ have frequencies $2, 3, 1$, respectively. Thus $n = 6$; the midpoints are $3, 5, 7$ kg.

$$\bar{x} \approx \frac{2(3) + 3(5) + 1(7)}{6} = \frac{28}{6} = \frac{14}{3} \approx 4.667 \text{ kg}.$$

The weighted squared-deviation sum is

$$2\left(3 - \frac{14}{3}\right)^2 + 3\left(5 - \frac{14}{3}\right)^2 + \left(7 - \frac{14}{3}\right)^2 = \frac{50 + 3 + 49}{9} = \frac{34}{3}.$$

$$s^2 \approx \frac{34/3}{6-1} = \frac{34}{15} \approx 2.267 \text{ kg}^2, \qquad s \approx \sqrt{\frac{34}{15}} \approx 1.506 \text{ kg}.$$

C7.4(b) – Comparison and limitation **[2]**

The estimated sample mean is $5 - 14/3 = 1/3 \approx 0.333$ kg below the model mean. The estimated variance is $34/15 - 2 = 4/15 \approx 0.267$ kg^2 above the model variance. The histogram fixes the frequencies exactly but does not reveal individual masses; replacing each mass by a midpoint introduces approximation into the sample statistics.



4 Summarized Solutions

C5

C5.1. (a) $E(A) = 10$ min, $Var(A) = 12$ min^2, $\sigma_A \approx 3.464$ min. (b) $E(B) = 9$ min, $Var(B) = 6.5$ min^2, $\sigma_B \approx 2.550$ min. (c) B: shorter mean time by 1 min and less variability.

C5.2. (a) $E(I) = 8$ h, $E(D) = 5$ h; both variances 4.5 h^2 and standard deviations ≈ 2.121 h. (b) Opposite density slopes shift the means; reflection preserves variance.

C5.3. (a) $T \sim \text{Exp}(2/15)$ in minutes; $E(T) = 7.5$ min, $Var(T) = 56.25$ min^2, $\sigma_T = 7.5$ min. (b) Dollar revenue: mean \$2400, variance 22500 dollars^2, standard deviation \$150; long-run daily average \$2400.

C6

C6.1. (a) $n = 20$, $k = 9$, $\widehat{P} = 0.45$. (b) $E = 0.05$; percentage error 10%; 5 percentage points below the model.

C6.2. (a) $n = 100$, $k = 72$, $\widehat{P} = 0.72$. (b) $E = 0.03$; complement 0.28; 3 percentage points below the model.

C6.3. (a) A: $n = 20$, $\widehat{P} = 0.70$, $E = 0.10$. B: $n = 200$, $\widehat{P} = 0.77$, $E = 0.03$. (b) B is closer; a larger sample does not guarantee smaller error.

C6.4. (a) Frequencies $(6, 12, 9, 3)$, $n = 30$, $\widehat{P} = 0.70$, $E = 0.05$, percentage error $\approx 6.67\%$. (b) At upper boundaries $(3, 6, 9, 12)$: cumulative frequencies $(6, 18, 27, 30)$ and probabilities $(0.20, 0.60, 0.90, 1)$; $\widehat{P}(X \geq 9) = 0.10$.

C7

C7.1. (a) $n = 5$, $\bar{x} = 12$, $s^2 = 5$, $s \approx 2.236$ (mean and standard deviation in degrees Celsius, variance in square degrees). (b) Mean equals μ; variance is 1 above σ^2; sample denominator 4.

C7.2. (a) $n = 6$, $\bar{x} = 5.5$ items, $s^2 = 5.1$ items^2, $s \approx 2.258$ items. (b) Mean 0.5 items above μ; variance 0.1 items^2 above σ^2.

C7.3. (a) Midpoints $(12, 16, 20)$ min; $n = 6$; midpoint estimates $\bar{x} \approx 17.333$ min, $s^2 \approx 10.667$ min^2, $s \approx 3.266$ min. (b) Mean ≈ 0.667 min below μ; variance ≈ 9.333 min^2 below σ^2; individual observations replaced by midpoints.

C7.4. (a) Frequencies $(2, 3, 1)$, $n = 6$, midpoints $(3, 5, 7)$ kg; midpoint estimates $\bar{x} \approx 4.667$ kg, $s^2 \approx 2.267$ kg^2, $s \approx 1.506$ kg. (b) Mean ≈ 0.333 kg below μ; variance ≈ 0.267 kg^2 above σ^2; individual observations replaced by midpoints.